The Market Alpha Engine, Explained Simply

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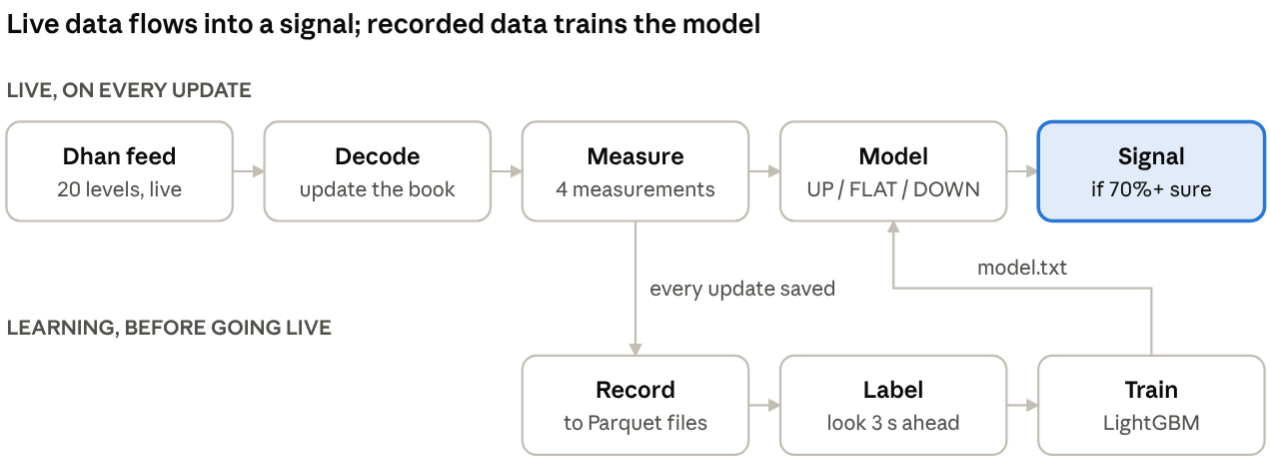
The big idea

We want a computer program that watches a stock's live buy and sell orders and guesses where its price will go in the next 3 seconds: up, down or nowhere.

Prices on a stock exchange do not move at random every instant. Just before a price rises, buyers often pile up and sellers thin out. A careful human could spot this, but it happens hundreds of times a minute, far too fast for anyone to follow.

So the program does the watching. It reads the market's live order data, turns it into a few simple measurements, and uses a trained prediction model to say "likely up" or "likely down". It only speaks up when it is at least 70% sure.

In finance, an edge like this is called **alpha**, which is why the project is called an *Alpha Engine*.



the whole system at a glance · live path and learning path

The top row runs on every market update. The bottom row runs first, on recorded days, and produces the model the top row uses.

What an order book is

An order book is the exchange's live list of everyone waiting to buy or sell a stock, and at what price.

Think of a busy vegetable market. Buyers shout what they will pay: "I'll pay ₹99!" Sellers shout what they want: "I'll sell at ₹101!" Nobody trades until a buyer and a seller agree on a price.

- A buyer's offer is called a **bid**. The highest bid is the **best bid**.
- A seller's asking price is called an **ask**. The lowest ask is the **best ask**.
- Each price level shows how many shares are waiting there (**quantity**) and how many separate people placed them (**orders**).

Our data source shows the best 20 price levels on each side. Here are the top 5 of an illustrative book (made-up numbers):

Buy orders	Shares wanted	Bid price (₹)	Ask price (₹)	Shares offered	Sell orders
4	286	1,508.05	1,508.15	161	3
17	119	1,507.95	1,508.25	167	7
4	430	1,507.90	1,508.35	65	1
6	752	1,507.65	1,508.45	247	7
8	883	1,507.60	1,508.50	221	2

The top row is where the action is: the best buyer and the best seller. Prices get worse for the trader the further down you go.

Where the data comes from

The order book arrives live from **Dhan**, an Indian stockbroker, over a connection that stays open all day and pushes every change as it happens.

It works like a live radio channel rather than a website you refresh:

1. **Tune in.** The program connects using your Dhan account details (client ID and access token).
2. **Choose stations.** It tells Dhan which stocks to watch, up to 50 at once.
3. **Listen.** Dhan then sends a fresh copy of the buy side and the sell side every time either one changes.

To save time, Dhan sends this data in a compact computer format, not readable text. Each update is exactly 332 bytes: a short label saying which stock and which side, then 20 price levels. The first job of the program is to decode these into numbers.

Connections drop sometimes. When that happens the program quietly reconnects and picks up where it left off. If Dhan says the login itself is wrong or expired, it stops and tells you, because retrying would not help.

The four measurements

Every time the book changes, the program boils it down to four simple measurements. Each one answers a plain question about the crowd of buyers and sellers. The examples use the made-up book shown earlier.

Measurement	The question it answers	How it is worked out	In the example
Spread	How far apart are the best buyer and the best seller?	Best ask − best bid	₹1,508.15 − ₹1,508.05 = **₹0.10**
Micro-price	Where is the "fair" price, given who is pushing harder at the top?	The midpoint, nudged toward the side with fewer shares waiting	**₹1,508.114**, a little above the midpoint of ₹1,508.10
Order book imbalance	Are there more shares waiting to be bought or to be sold?	(buy shares − sell shares) ÷ (buy shares + sell shares), top 5 levels	(2,470 − 861) ÷ 3,331 = **+0.48**
Bid and ask spectrum	How much is waiting close to the current price, and how much far away?	Shares within 0.05%, 0.05–0.15% and 0.15–0.50% of the midpoint	All 2,470 top-5 buy shares sit within 0.05% (about ₹0.75)

What each one hints at:

- **Spread.** A narrow spread means a calm, busy market. A wide one means traders are unsure.
- **Micro-price.** Here 286 shares want to buy at the top but only 161 are offered, so the sellers' side is likely to be used up first. The fair price leans upward.
- **Imbalance.** It runs from −1 (only sellers) to +1 (only buyers). At +0.48, buyers clearly outnumber sellers, which often comes before a small rise.
- **Spectrum.** It shows *where* the crowd stands. Lots of buyers right next to the price form a strong floor; a thin floor can give way quickly.

No single measurement predicts the price. The point is to give the model a clear, numeric picture of the book's shape.

Teaching the computer

Before it can predict anything, the model must learn from real history: thousands of past moments, each paired with what the price actually did next.

Step 1: Record. During market hours the program saves every set of measurements, with the time and the price, to files on disk. It uses a compact file format called Parquet because a single day can produce hundreds of thousands of rows.

Step 2: Write the answer key. For every saved moment, the program looks at the price 3 seconds later and gives that moment a label:

Price 3 seconds later	Label	Example, starting at ₹1,500.00
More than 0.01% higher	UP	above ₹1,500.15
More than 0.01% lower	DOWN	below ₹1,499.85
Anything in between	FLAT	₹1,499.85 to ₹1,500.15

Step 3: Train. A model called **LightGBM** studies the measurements and the labels together. It builds many small yes/no rules, such as "if buyers outnumber sellers by a lot and the spread is narrow, UP is likely". Each new rule fixes mistakes the earlier ones made. It is told to pay equal attention to all three outcomes, even if one is rarer.

Step 4: Test fairly. The model is tested only on moments that come *after* everything it learned from, for example learning from 9:15 AM to 1:00 PM and testing from 1:00 PM to 3:30 PM.

This is like a student who practises on last year's papers and is then tested on a paper they have never seen. If we shuffled the moments randomly instead, the test would contain moments a split second away from ones it trained on. That is like seeing the answers before the exam: the score would look great and mean nothing.

The finished model is saved to a file called `model.txt`, ready for live use.

Using it live

Once trained, the model runs on every single update from the market, and raises a flag only when it is at least 70% sure.

For each update, in order:

1. Dhan sends a new snapshot of the buy or sell side.
2. The program decodes it and refreshes its copy of the order book.
3. It recalculates the four measurements.

4. The model returns three chances that add up to 100%: UP, FLAT and DOWN.
5. It turns those chances into a signal:

If the model says	Signal	Meaning
Chance of UP is above 70%	**LONG**	The price will probably rise soon
Chance of DOWN is above 70%	**SHORT**	The price will probably fall soon
Neither is above 70%	**FLAT**	Not sure enough; stay quiet

Speed matters. A pattern that lasts a few seconds is useless if the warning arrives late. The goal is for all five steps to finish in under one millisecond, a thousandth of a second, roughly a hundred times faster than a blink.

The signal is a hint, not a trade. The program does not buy or sell anything by itself.

What you see on screen

While it runs, the program shows a live dashboard in the terminal window that refreshes several times a second.

Panel	What it shows
Header	Which stock is being watched, whether the connection is live, and the time
Top 5 depth	The five best buy and sell levels, like the table in "What an order book is"
Microstructure	The current spread, micro-price, imbalance and spectrum
Model	Bars for the chances of UP, FLAT and DOWN, the current signal, and recent signals
Footer	Updates per second, how fast each step ran, and how many rows have been recorded
Events	Recent messages, such as a reconnection or a new signal

What this is not

This is a research tool that makes educated guesses. It is not a guaranteed money-maker and not financial advice.

- **Patterns can be weak or fade.** What worked last week may stop working, and other traders hunt for the same patterns.
- **Small moves, real costs.** A 0.01% move on a ₹1,500 share is only ₹0.15. Brokerage, taxes and the spread can easily cost more than that.
- **It needs real data first.** The pipeline has been tested on made-up market data, which shows the machinery works. Only several days of recorded real data can show whether a useful pattern exists.
- **Others are faster.** Professional trading firms place their computers inside the exchange building. A home internet connection adds delays this program cannot remove.
- **It does not trade.** It only shows signals. Anyone acting on them takes on the full risk.

Glossary

Term	Plain meaning
Alpha	An edge: a way to predict prices a little better than chance
Ask	The price a seller wants; the best ask is the lowest one
Bid	The price a buyer offers; the best bid is the highest one
Dhan	The Indian stockbroker whose live data feed this project uses
FLAT	No strong view; the price is expected to stay about the same
High-frequency	Reacting to market changes many times per second
Imbalance	Whether more shares are waiting to be bought or to be sold, from −1 to +1
Label	The right answer for a past moment: UP, DOWN or FLAT

Term	Plain meaning
LightGBM	The type of prediction model used; it learns many small yes/no rules
LONG	Signal that the price will probably rise
Micro-price	A "fair" price that leans toward the side under more pressure
Millisecond	One thousandth of a second
Order book	The live list of all waiting buy and sell orders for a stock
Parquet	A compact, fast file format for storing large tables
SHORT	Signal that the price will probably fall
Spectrum	How many shares wait at different distances from the current price
Spread	The gap between the best ask and the best bid